

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Final Examination of semester 1

Year: 2014

Subject: 392133 Physics III

Section: 15-16

Date: 12 December 2014

Time: 08.00-10.00

Name.....ID:.....Class:.....

Instructions

1. Cheating will result in failure of all classes registered for the current semester.
Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken inside the examination room.
3. Express your answer in English. Other languages will be discarded.
4. A scientific calculator is allowed.
5. Electronic communication devices are NOT allowed in the examination room.
6. The examination has 7 page (including this page), 6 problems and a total scores of 60 points.
7. Write all your answers on this examination sheet.

1.1 [3 points] The index of refraction for red light in water is 1.331 and for blue light is 1.340. If a ray of white light enters the water at an angle of incidence of 83.00° , what are the underwater angles of refraction for the blue and red components of the light?

1.2 [3 points] The laws of refraction and reflection are the same for sound as for light. The speed of sound in air is 340 m/s and in water it is 1510 m/s. If a sound wave approaches a plane water surface at an angle of incidence of 12.0° , what is the angle of refraction?

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1.3 [4 points] A layer of kerosene ($n = 1.45$) is floating on water ($n = 1.33$). For what angles of incidence at the kerosene-water interface will light be totally internally reflected within the kerosene?

2.1 [2 points] A man 2 m tall looks at himself in a full-length mirror (floor-to-ceiling). Where in the mirror must he look to see his feet?

2.2 [3 points] A concave mirror has a focal length of 20 cm. What is the position (in cm) of the object if the image is upright and is two times larger than the object? In addition please draw a ray diagram for the conclusion.

2.3 [5 points] Show that the image equation of a thin lens can be written by $\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$ where f is the focal length, d_o is the object distance and d_i is the image distance.

3.1 [3 points] A contact lens is made of plastic with an index of refraction of 1.50. The lens has an outer radius of curvature of +2.0 cm and an inner radius of curvature of +2.5 cm. What are the focal length and the power of the lens?

3.2 [5 points] A compound microscope is made with an objective lens ($f_o = 0.90$ cm) and an eyepiece ($f_e = 1.1$ cm). The lenses are separated by a distance of 10 cm. If an object is 1.0 cm in front of the objective lens, where will the final image of the eyepiece be located?

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3.3 [2 points] A far-sighted student has a near point of 1.0 m. Calculate the focal length (in cm) of the glasses needed so the near point will be normal (25 cm).

4.1 [5 points] A 10-cm focal length converging lens is used to magnify small newspaper print 2.0 mm high. Calculate the height of the image (in mm) for the maximum magnification for a normal eye. In addition, please sketch (approximately) a ray diagram and derive the equation used in your calculation.

4.2 [5 points] A telescope is constructed with two lenses separated by a distance of 25 cm. The focal length of the objective is 20 cm. The focal length of the eyepiece is 5 cm. Calculate the magnitude of the angular magnification of the telescope. In addition, please sketch (approximately) a ray diagram to confirm your result.

5.1 [4 points] In a double slit experiment, the distance between the slits is 0.2 mm and the distance to the screen is 100 cm. What is the addition result between the waves from the two slits arriving at a point 5 mm from the central maximum when the wavelength is 400 nm? Explain your findings from your calculation.

5.2 [3 points] Plane waves of ultrasound, $f = 51.0$ kHz, impinge on a flat plane with a 1.00 cm-wide slit. What is the total angular width (2θ) of diffracted sound waves when the speed of sound in air is 340 m/sec.

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5.3 [3 points] White light is spread out into spectral hues by a diffraction grating. If the grating has 1,000 lines per cm, at what angle will red light ($\lambda = 640$ nm) appear in first order?

6.1 [3 points] Monochromatic light ($\lambda = 500 \text{ nm}$) is incident on a soap bubble ($n = 1.40$). How thick is the bubble (in nm) if destructive interference occurs in the reflected light?

6.2 [4 points] Unpolarized light is passed through three successive Polaroid filters, each with its transmission axis at 45.0° to the preceding filter. What percentage of light gets through?

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6.3 [3 points] Sunlight reflected from a smooth ice surface is completely polarized. Determine the angle of incidence. ($n_{\text{ice}} = 1.31$.)